Data for the modified IEEE 6-bus system

1. Topology

The topology of modified IEEE 6-bus system is shown in Fig. 1

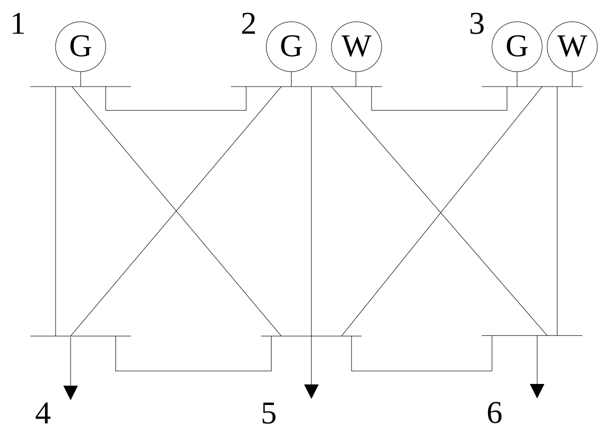

Fig. 1. Topology for the modified IEEE 6-bus system

The traditional units are at bus 1, 2 and 3. The loads are at bus 4, 5, and 6. The wind power is injected at bus 2 and 3. The swing bus is bus 1.

2. Data

The network data for the loads and transmission lines are the same with standard IEEE 6-bus system. The generator cost data are revised and shown in Table I, the load shedding cost data are shown in Table II, the wind spillage cost is 1 $/MW-hr.

TABLE I
GENERATOR COST DATA

	Stage 1	**Stage 2**	**Stage 3**
Generator @ bus1	10 $/MW-hr	20 $/MW-hr	30 $/MW-hr
Generator @ bus2	15 $/MW-hr	25 $/MW-hr	35 $/MW-hr
Generator @ bus3	20 $/MW-hr	30 $/MW-hr	40 $/MW-hr

TABLE II
LOAD SHEDDING COST DATA

Load @ bus4	**Load @ bus5**	**Load @ bus6**
90 $/MW-hr	100 $/MW-hr	100 $/MW-hr

Data for the Modified Gansu power grid in China

1. Topology

The topology of the Modified Gansu power grid in China is shown in Fig. 2.

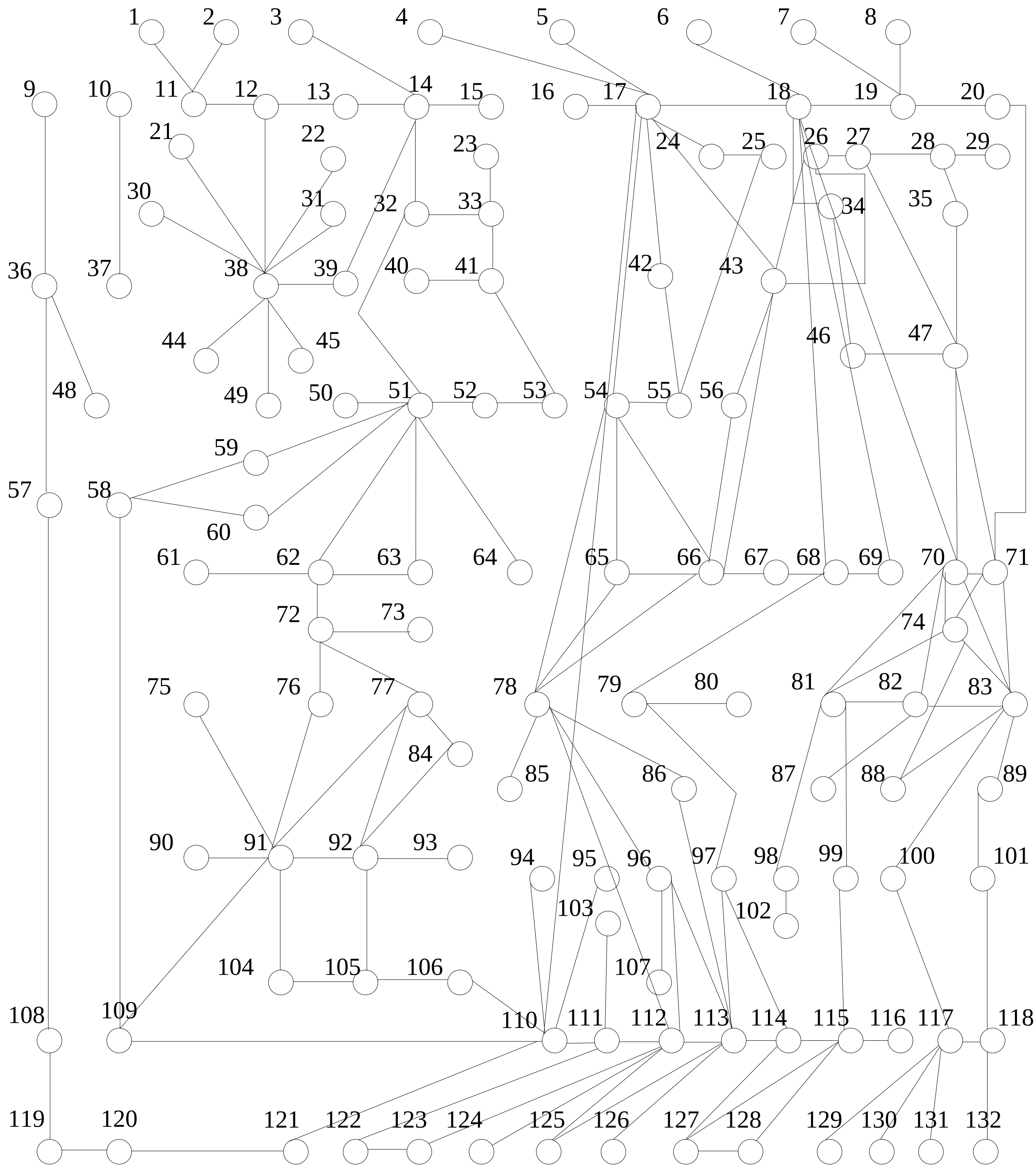

Fig. 2 Topology for the Gansu power grid in China

2. Data

There are 132 buses and 244 transmission lines in this system. There are 25 equivalent traditional generators and 6 wind power generators, injected in bus 14, 38, 51, 63, 54 and 58. The data for loads, transmission lines, generators and generator cost are given in Table III-Table VI respectively.

TABLE III
LOAD DATA

Bus No.	Load(MW)		Bus No.	Load(MW)		Bus No.	Load(MW)
1	0		40	0		79	1.9165
2	0		41	20.2357		80	0.5589
3	0		42	0		81	12.6094
4	100.1029		43	29.2755		82	21.3294
5	27.9391		44	0		83	5.3237
6	7.111		45	0		84	0
7	0		46	0		85	0.6479
8	0.2465		47	0		86	0
9	0		48	0		87	0
10	74.99		49	0.183		88	1.5052
11	0		50	0		89	1.5822
12	0		51	30.3063		90	0
13	0		52	0.0554		91	26.332
14	0		53	7.422		92	2.3161
15	0		54	5.8447		93	0
16	7.0242		55	6.9477		94	22.8798
17	0		56	1.8794		95	23.9808
18	0		57	4.9793		96	2.6539
19	17.7797		58	0		97	2.94
20	100.0248		59	0		98	0.5148
21	0		60	0		99	6.7314
22	0.0051		61	0		100	0
23	0		62	20.0952		101	0
24	2.8426		63	0.3619		102	0.2794
25	0		64	0		103	0
26	0		65	0.7896		104	3.532
27	1.3985		66	7.3182		105	3.18
28	2.4302		67	2.2274		106	7.8261
29	0		68	7.4661		107	0
30	0		69	25.2572		108	0
31	0		70	0		109	4.3852
32	1.929		71	6.0774		110	0
33	7.7999		72	1.9517		111	12.0218
34	9.5451		73	0.0167		112	4.1494
35	1.5348		74	7.1316		113	0
36	24.6856		75	0		114	0
37	1.0253		76	2.5747		115	5.67
38	11.7182		77	3.8423		116	0
39	0.5664		78	2.4201		117	0

TABLE III
LOAD DATA (CONTINUED)

Bus No.	Load(MW)		Bus No.	Load(MW)		Bus No.	Load(MW)
118	3.7815		123	0		128	1.4446
119	0		124	0		129	0
120	12.9078		125	0.5935		130	0
121	32.0279		126	0		131	0.3733
122	0		127	1.4833		132	0.0355

TABLE IV
BRANCH DATA

From bus No.	To bus No.	x	Thermal limit (MW)		From bus No.	To bus No.	x	Thermal limit (MW)
1	12	0.47	80		17	19	0.15	100
2	12	0.107	80		17	66	0.15	100
3	14	0.146	80		17	68	0.15	100
4	17	0.3	100		17	77	0.15	100
4	14	0.3	100		17	73	0.15	100
4	42	0.3	100		17	26	0.15	100
4	15	0.3	100		17	24	0.158	80
4	38	0.3	100		17	5	0.12	80
4	13	0.3	100		17	110	0.15	100
4	71	0.3	100		17	43	0.13	80
4	16	0.3	100		17	55	0.179	80
4	53	0.3	100		17	54	0.1	80
4	66	0.3	100		18	70	0.12	100
4	77	0.3	100		18	20	0.12	100
4	18	0.3	100		18	19	0.1	100
6	18	0.3	100		19	20	0.14	100
7	19	0.3	100		20	71	0.15	100
8	19	0.3	100		20	70	0.15	100
9	36	0.13	100		22	38	0.15	100
9	10	0.13	100		23	33	0.1	80
10	37	0.5	100		27	28	0.143	80
10	36	0.5	100		29	28	0.143	80
10	38	0.5	100		29	35	0.143	80
10	32	0.5	100		30	38	0.95	80
10	17	0.4	100		31	38	0.95	80
10	21	0.4	100		31	66	0.95	80
10	39	0.4	100		32	14	0.219	80
11	12	0.5	80		33	32	0.1	80
12	38	0.2	80		33	25	0.1	80
12	13	0.3	80		33	39	0.1	80
12	17	0.4	80		34	18	0.46	80
12	39	0.4	80		35	28	0.219	80
14	15	0.5	80		35	26	0.219	80
16	17	0.1	80		35	43	0.219	80
17	18	0.15	100		35	69	0.219	80
17	13	0.15	100		36	37	0.1	100
17	53	0.15	100		36	57	0.13	100

TABLE IV
BRANCH DATA (CONTINUED 1)

From bus No.	To bus No.	x	Thermal limit (MW)		From bus No.	To bus No.	x	Thermal limit (MW)
36	39	0.13	100		51	32	0.35	80
36	48	0.13	80		51	54	0.35	80
37	58	0.13	100		51	63	0.326	80
37	49	0.13	100		51	50	0.35	80
37	53	0.13	100		51	52	0.47	80
37	38	0.12	80		51	39	0.35	80
38	39	0.11	80		51	72	0.35	80
38	13	0.11	80		53	52	0.4	80
38	66	0.11	80		53	4	0.4	80
38	49	0.7	80		55	25	0.265	80
38	53	0.7	80		55	54	0.193	80
38	30	0.113	80		56	43	0.56	80
38	21	0.113	80		56	71	0.56	80
38	22	0.1	80		56	66	0.36	80
38	72	0.1	80		57	108	0.19	100
38	44	0.25	80		58	109	0.15	100
38	45	0.105	80		59	51	0.15	100
38	49	0.146	80		59	58	0.15	100
40	41	0.5	80		60	51	0.15	100
41	53	0.12	80		60	58	0.15	100
41	51	0.12	80		61	62	0.8	80
41	54	0.12	80		62	72	0.1	80
41	42	0.12	80		62	61	0.48	80
42	55	0.12	80		63	62	0.208	80
42	53	0.12	80		63	65	0.208	80
43	66	0.171	80		64	51	0.1	80
43	68	0.171	80		65	54	0.118	80
44	38	0.171	80		65	66	0.63	80
45	38	0.171	80		65	78	0.106	80
46	47	0.22	80		65	78	0.106	80
46	70	0.22	80		65	71	0.106	80
47	35	0.176	80		59	51	0.15	100
47	83	0.521	80		59	58	0.15	100
47	27	0.221	80		60	51	0.15	100
47	70	0.244	80		60	58	0.15	100
48	36	0.221	80		61	62	0.8	80
50	51	0.6	80		62	72	0.1	80

TABLE IV
BRANCH DATA (CONTINUED 2)

From bus No.	To bus No.	x	Thermal limit (MW)	From bus No.	To bus No.	x	Thermal limit (MW)
62	61	0.48	80	78	81	0.1	80
63	62	0.208	80	78	85	0.4	80
63	65	0.208	80	78	84	0.108	80
64	51	0.1	80	78	112	0.108	80
65	54	0.118	80	79	68	0.39	80
65	66	0.63	80	79	80	0.44	80
65	78	0.106	80	79	97	0.35	80
65	78	0.106	80	81	98	0.251	80
65	71	0.106	80	81	70	0.5	80
62	61	0.48	80	81	97	0.5	80
63	62	0.208	80	81	101	0.5	80
63	65	0.208	80	82	87	0.5	80
64	51	0.1	80	82	70	0.25	80
65	54	0.118	80	83	74	0.548	80
65	66	0.63	80	83	100	0.548	80
65	78	0.106	80	83	89	0.11	80
65	78	0.106	80	83	88	0.5	80
65	71	0.106	80	83	70	0.153	80
65	53	0.106	80	84	92	0.15	80
67	68	0.5	80	84	68	0.15	80
67	66	0.7	80	86	113	0.13	80
68	18	0.13	80	91	77	0.15	80
68	96	0.41	80	89	101	0.548	80
69	18	0.21	80	91	92	0.25	80
69	78	0.31	80	91	66	0.25	80
70	71	0.21	100	91	38	0.25	80
70	110	0.1	100	91	90	0.25	80
72	73	0.24	80	91	104	0.1	80
72	76	0.1	80	93	92	0.2	80
74	70	0.112	80	96	107	0.4	80
75	91	0.1	80	98	102	0.126	80
76	91	0.25	80	99	81	0.42	80
77	84	0.25	80	100	117	0.7	80
78	86	0.58	80	101	118	0.4	80
78	121	0.58	80	101	115	0.2	80
78	96	0.1	80	102	98	0.2	80
78	71	0.1	80	103	111	0.2	80

TABLE IV
BRANCH DATA (CONTINUED 3)

From bus No.	To bus No.	x	Thermal limit (MW)		From bus No.	To bus No.	x	Thermal limit (MW)
104	105	0.1	80		114	97	0.12	80
104	121	0.1	80		115	99	0.12	80
104	73	0.1	80		115	116	0.154	80
106	105	0.1	80		115	97	0.154	80
106	110	0.1	80		116	115	0.12	80
107	96	0.1	80		118	132	0.76	80
108	119	0.13	100		118	113	0.76	80
109	110	0.125	100		118	97	0.76	80
110	111	0.6	80		118	117	0.379	80
110	95	0.25	80		119	120	0.11	80
110	121	0.12	80		119	110	0.11	80
110	94	0.24	80		120	121	0.17	100
112	113	0.194	80		122	123	0.17	100
112	96	0.163	80		122	111	0.17	100
112	111	0.25	80		123	112	0.17	100
112	97	0.25	80		124	112	0.17	100
113	96	0.71	80		126	113	0.17	100
113	126	0.93	80		127	128	0.246	80
113	97	0.1	80		127	114	0.142	80
113	125	0.267	80		127	115	0.191	80
114	115	0.1	80		129	117	0.15	80
114	113	0.4	80		130	117	0.25	80
131	117	0.56	80		N/A	N/A	N/A	N/A

TABLE V
GENERATOR DATA

Bus No.	P_{max} (MW)	P_{min} (MW)		Bus No.	P_{max} (MW)	P_{min} (MW)		Bus No.	P_{max} (MW)	P_{min} (MW)
4	180	20		43	92	0		87	100	10
6	180	20		50	85	5		90	105	10
7	87	5		51	95	5		91	110	10
8	87	5		46	110	10		114	80	5
10	220	20		61	125	10		121	80	5
16	92	5		69	96	5		131	80	5
20	70	5		73	82	5		132	80	5
23	83	5		80	82	5		N/A	N/A	N/A
40	67	5		85	92	0		N/A	N/A	N/A

TABLE VI
GENERATOR COST DATA

Bus No.	Stage 1 ($/MW-hr)	Stage 2 ($/MW-hr)	Stage 3 ($/MW-hr)		Bus No.	Stage 1 ($/MW-hr)	Stage 2 ($/MW-hr)	Stage 3 ($/MW-hr)
4	5	6	7		46	3	5	6
6	4	7	8		61	3	5	6
7	3	6	9		69	5	8	9
8	4	5	6		73	5	6	7
10	4	8	9		80	2	6	7
16	2	8	9		85	4	6	8
20	6	7	8		87	5	8	9
23	8	9	10		90	3	5	7
40	1	9	10		91	2	8	9
132	6	8	9		N/A	N/A	N/A	N/A